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The Evolution of Tetrapod Ears and the Fossil Record

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Tetrapod
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Evolution
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Abstract
In the earliest tetrapods, the fenestra vestibuli was a large hole in the braincase wall bounded by bones of different embryological origins: the otic capsule and occipital arch components, and also, in all except the Devonian *Acanthostega*, the dermal parasphenoid. This means that the hole lay along the line of the embryonic metotic fissure. Early tetrapod braincases were poorly ossified internally, and no specialized opening for a perilymphatic duct is evident. It is arguable that the earliest tetrapods had neither a perilymphatic duct crossing the otic capsule nor a specialized auditory receptor in a separate lagenar pouch. The primitive tetrapod condition is found in the earliest amniotes, and the separate development of (1) a fenestra vestibuli confined to the limits of the otic capsule, (2) a specialized pressure relief window also derived from components on the line of the metotic fissure, (3) a nonstructural, vibratory stapes and (4) increased internal ossification of the internal walls of the otic capsule, can be traced separately in synapsids, lepidosauromorph diapsids, archosauromorph diapsids, probably turtles, and amphibians. This suggests separate development of true tympanic ears in each of these groups. Developments indicating the existence of a true tympanic ear in amniotes are first found in animals from the Triassic period, and a correlation with the evolution of insect sound production is suggested.

Introduction

In the late 1970s, the generally accepted view of the evolution of tetrapod tympanic ears derived substantially from that of Goodrich [1930]. They were considered to be homologues resulting from a single common ancestry among the earliest forms. Anthracosaurs, part of the old 'labyrinthodont' group, were considered to be related to the origin of amniotes, and their stapes were therefore thought to hold the key to the evolution of amniote ears. However, their stapes, though often reconstructed [e.g., Thomson, 1966] had never been found. When I discovered two of them, in two closely related anthracosaurs from the Carboniferous of Yorkshire [Clack, 1983], I was faced with interpreting something quite unexpected and unlike the rodlike structure they were assumed to have been. Instead, each had an irregular footplate, narrow neck and flattened, fan-shaped distal portion. The timing of my discovery was fortuitous in that it coincided with a number of studies both in palaeontology and neontology which combined to upset the conventional view and replace it with one in which a 'tympanic ear' evolved separately in several different groups [Wever,

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1978; Lombard and Bolt, 1979; Carroll, 1980; Smithson, 1982]. Since then, work on both extant and fossil forms has only served to strengthen this revised view [Carr, 1992; Manley and Gleich, 1992; Laurin and Reisz, 1997].

In more recent years, I have been working on the Upper Devonian tetrapod *Acanthostega gunnari* from Greenland. Exceptional material from this animal has provided much new and even revolutionary information about early tetrapods. Propitiously, this material included the braincase and stapes, which have not only provided ideas about the early evolution of tetrapod braincases but have inspired me to study the evolution of the ear region in tetrapods in some detail [Clack, 1989, 1992, 1994, 1997]. The whole story has turned out to be much more complicated, and much more interesting, than the simple one I learned as a student many years ago.

Beginning with the fenestra vestibuli and its relationship to the stapes in *Acanthostega*, I focused next on this region in other tetrapods [Clack, 1997]. This led me to review the evolution of other parts of the ear apparatus to document and analyze the evolution of the ear throughout the tetrapods. By studying which, when and how many times various components arose, we actually learn a great deal more than simply whether a tympanic ear was present or not. This paper is a brief overview of the story as it is revealed in the fossil record, with some thoughts about how it may be refined in the future by input from further fossil finds, from phylogenetic analysis, and from neontological studies.

Otic Characters in the Fossil Record

A number of bony characters lend themselves to preservation as fossils, revealing information about middle and inner ear structure both directly and indirectly. The stapes is an obvious candidate, and fossils reveal its size, shape, orientation, presence or absence of a stapedial foramen or a dorsal process, and relationships to the bones of the skull and to the ossifications of the braincase. Many analyses in the past have relied on descriptions such as 'large', 'small', 'rodlike', 'platelike', and dorsally or ventrally directed, to infer its contribution to a hearing mechanism, but these are often subjective and cannot be relied on in isolation. However, taken in combination with other information, inferences may be drawn about ear function.

The 'otic notch' of older literature covers features to which several bones contribute, but these may be different bones in different groups. The term has also been applied to features that are more readily interpreted in other ways. However, the shape and configuration of the quadrate and its relationship to the braincase, palate, quadratojugal, squamosal and other bones of the skull roof should each be considered. Together they may inform us about the possible presence of a tympanic membrane and the development of the middle ear cavity, but also the extent of any kinesis between the skull, braincase, and roofing bones may have implications for stapedial function.

Several properties of the otic capsule can be tracked through the fossil record. The first of these is the size, position, and relationship of the fenestra vestibuli (FV), often referred to as the fenestra ovalis or oval window. This opening provides the interface between the middle ear and the inner ear. (For this reason it seems to get neglected in anatomical accounts of one or the other, 'falling between two stools', as it were.) It emerges that the way in which different braincase elements contribute to the margin of this fenestra is highly significant. Internal to the otic capsule, the ossifications surrounding the semicircular canals, the saccular, lagenar, and cochlear pouches, and the degree to which the medial wall is ossified all provide important clues. Finally, the relationships of the otic capsule to the occipital arch components tell us a great deal about the development of a dedicated pressure relief window (PRW), known as the fenestra rotunda (FR) or round window in mammals and given this or other names (such as fenestra pseudorotunda) in other amniotes. This structure is almost always associated with acute terrestrial hearing and, like the formation of the FV, turns out to be a key feature in understanding the evolution of ears in the fossil record.

Taking all these characters into account, there is a wealth of information in the fossil record about the evolution of the ear region. Space prevents a comprehensive review here, but this is planned by the author for the future. Meanwhile this paper presents some key points and a summary of data, ideas, and directions to be followed up.

The Otic Region of Stem-Tetrapods, and the Origin of the Fenestra Vestibuli

There is now a fairly clear pattern emerging showing the basal conditions for the otic region from which all later tetrapods have diverged. Stapes are now known for a number of very early tetrapods, including *Acanthostega* [Clack, 1989], *Whatcheeria* [Lombard and Bolt, 1995], loxommatids [Beaumont and Smithson, 1997], anthracosaurs [Clack, 1983] and *Greererpeton* [Carroll, 1980]. In each case, the bone is flattened in cross-section and expanded distally (fig. 1). They are associated with ossifications of the palatoquadrate and/or dermal pterygoid which show no specializa-

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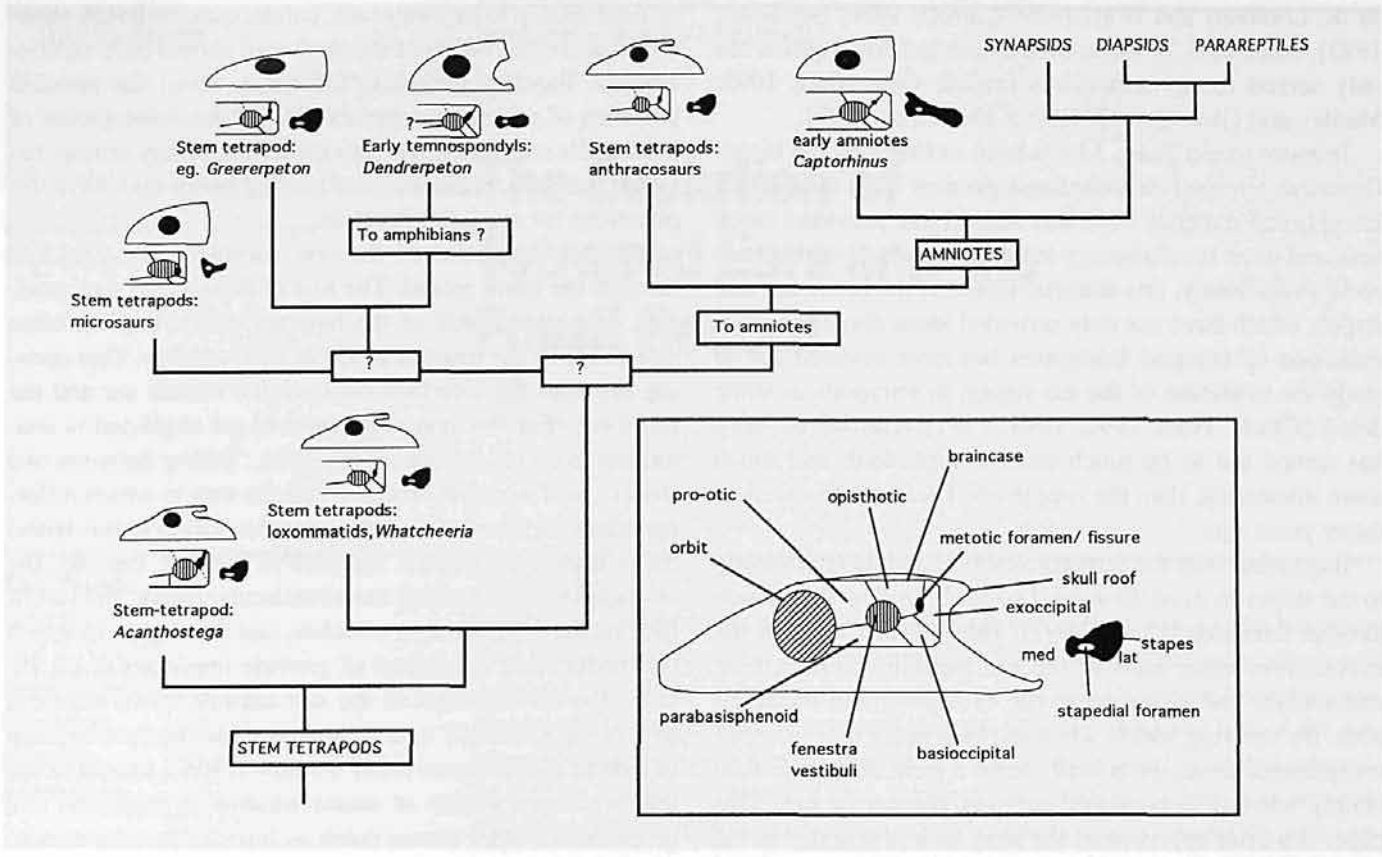


Fig. 1. Cladogram with icons of early tetrapods showing characters of the otic region. Inset diagram explains the construction of the icons illustrating these regions in this and figures 3, 5.

tions suggestive of an inflated middle ear cavity. The inference is that this was the primitive condition among early tetrapods.

As with all later tetrapods, the head of the stapes in *Acanthostega* was incorporated into the braincase wall [Clack, 1994]. In fishes, the hyomandibula is the equivalent bone to the stapes in tetrapods, and it articulates at facets on the otic capsule wall. The advent of the tetrapod condition forced some fundamental remodeling of this area, presumably affecting the course of such structures as the jugular vein, orbital artery and cranial nerve VII. However, other parts were left relatively unchanged. These included the occipital arch and the basioccipital and dorsal parts of the capsule containing the semicircular canals. The footplate of the stapes was thus brought into intimate proximity to the pars inferior of the inner ear, most significantly to the saccule. The events that brought about such a momentous rearrangement remain unknown, but the effects were profound and far reaching.

In *Acanthostega*, the footplate of the stapes occupies a large proportion of the braincase wall, including the equivalent of that feature of the braincase known as the vestibular fontanelle in fish such as *Eusthenopteron*. This unossified gap represents part of the metotic fissure which lies between the otic capsule and occipital arch in all vertebrate embryos. In many fossil fishes and in *Acanthostega* it remained unconsolidated in the adult. The vestibular fontanelle was bordered by the proötic and opisthotic dorsally and by the basioccipital ventrally [Jarvik, 1980]. The hole into which the stapes fits in tetrapods, the fenestra vestibuli (see inset diagram in fig. 1), shares the same relationships to the surrounding bones as does the vestibular fontanelle, though it also occupies the space which formerly bore the facets for the head of the hyomandibula. An important inference is that at its inception, the fenestra vestibuli was confluent with the metotic fissure and was not exclusively a feature of the otic capsule. Though in modern tetrapods the fenestra vestibuli usually forms within the confines of the

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capsule, sometimes by death of the cells forming the capsule wall [Crompton, 1953], tracing its fossil history shows that it had different relationships originally, at least in adult forms.

We now have good information about this region in a wide range of early tetrapods representing several independent groups. In some cases, there is also complementary information about the stapes and its relationships to the braincase and skull roof. These tell a consistent story about the origin, size, and relationships of the fenestra vestibuli, as follows.

In all the Palaeozoic taxa for which we have information, the fenestra vestibuli was a large hole whose margins were formed by contributions from both the otic capsule and the occipital arch. In *Acanthostega*, the parabasisphenoid was excluded by the basioccipital, as it was from the vestibular fontanelle in fishes, but in later taxa the dermal parasphenoid had lengthened posteriorly, such that the parasphenoid (and probably the basisphenoid internal to it) contributed the anteroventral parts of the margin (fig. 1). Stapedial footplates are known in several stem tetrapods: for example, in the microsaur *Rhynchonkos* (as *Goniorhynchus* [Carroll and Gaskill, 1978]) the colosteid *Greererpeton* [Carroll, 1980] and the amniotes *Captorhinus* [Heaton, 1979] and *Ophiacodon* [Romer and Price, 1940]. The footplate contacted four bones or their equivalent regions: proötic, opisthotic, basioccipital and parabasisphenoid. The exoccipital, dorsal to the basioccipital, is sometimes included in the fenestral margin or may be excluded by the opisthotic. The fossil record sends a clear signal that this multipartite state was the primitive condition for the fenestra vestibuli (fig. 1).

Another clear signal among early tetrapods is that the interior of the otic capsule was invariably poorly ossified. Contrasting with the condition in a number of Palaeozoic fishes, both actinopterygians and sarcopterygians, it may arguably represent a paedomorphic state in tetrapods. It may in turn be connected with the enlargement of the fenestra vestibuli and the incorporation of the stapedial footplate into the hole. Functionally, the contents of the otic capsule were not isolated by bone from the rest of the brain. Furthermore, the metotic fissure remained unsealed, with no obviously defined metotic foramen for the jugular vein, the cranial nerves IX, X, and XI, or the perilymphatic duct. In later tetrapods these leave through separate remnants of the gradually diminishing fissure. It is not clear that the earliest tetrapods even possessed a perilymphatic duct, given its absence in coelacanths and lungfish [Fritzsch, 1992], and its incongruent courses in amphibians and amniotes [Lombard and Bolt, 1979]. There is no evidence for it in the braincases of early tetrapods.

One of the main contentions of this paper is that lack of ossification in the ears of early tetrapods does not necessarily mean lack of data. On the contrary, the story of the evolution of the ear is one of increasing ossification and differentiation of this part of the braincase, particularly evident among amniotes, which, it is suggested, parallels a story of increasing specialization towards terrestrial hearing. The bulk of this paper outlines developments in various lineages, particularly amniotes, to the fenestra vestibuli and to the other components of the ear region. Figure 2 places the animals in their stratigraphic context.

Amniotes

Amniotes have a reasonably good fossil record, and though the interrelationships of the groups to each other are the subject of ongoing dispute, there is general agreement about which fossil animals belong to which groups. Of the amniote groups as a whole, the record of the mammalian lineage is the best represented, their internal relationships are best understood, and their otic regions have the best fossil record, which has been thoroughly and continuously investigated over many years [see Allin and Hopson, 1992, and references therein]. They will therefore be used as an exemplar before turning to less well represented groups.

The Synapsid-Mammal Lineage

Mammals are the crown group of a long radiation of amniotes known as the synapsids, the earliest recorded member being late Carboniferous (Pennsylvanian) in age [Reisz, 1972] (fig. 2). There is fairly general agreement that the synapsid lineage sprang from the base of the amniote radiation, so that mammals have a long history separate from other amniotes. The first true mammals emerged during the late Triassic period [Lucas and Luo, 1993], while therian mammals (marsupials plus placentals) had emerged by the late Cretaceous [Kemp, 1982; Kermack and Kermack, 1984].

The otic region in therians is characterized by a suite of unique characters, several of which are traceable in the fossil record, consequent upon the well-ossified bony capsule that closely surrounds the inner ear structures. These include the series of three otic ossicles, round (pressure relief) window and coiled cochlea. The earliest coiled cochlea is recorded from marsupial fossils from the late Cretaceous [Meng and Fox, 1995], while *Morganucodon*, one of the best-known Lower Jurassic forms [Kermack et al., 1981], still had a short and straight cochlea. Evolution of the cochlea is associated with development of a bony promontori-

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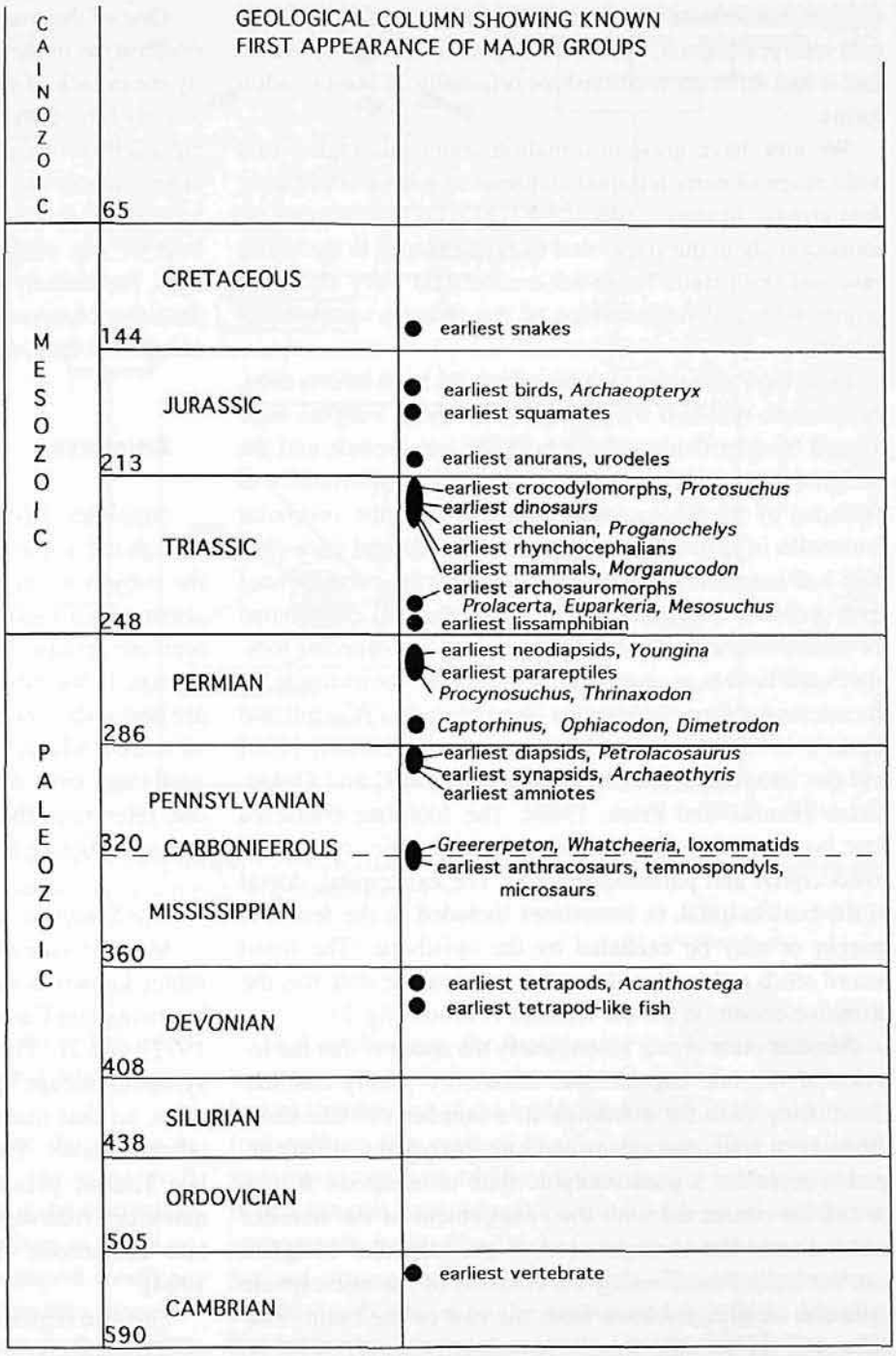


Fig. 2. Stratigraphic table placing the known first appearance of major groups and selected genera in their chronological context. Numbers indicate millions of years ago for the beginning of each period.

um, formed by the petrosal bone, a fusion of the otic capsule (proötic and opisthotic) bones [Luo et al., 1995]. The evolution of this structure too, can be tracked among the earliest mammals.

In *Morganucodon,* the fenestra vestibulae is separate from the round window, and both are situated within the otic capsule, while the jugular foramen (for the exit of the internal jugular vein and cranial nerves IX and X) is bordered in part by the otic capsule and in part by the basioccipital (fig. 3). Only the proximal part of the stapes of *Morganucodon* is known, but it is pierced by a large foramen, and the footplate is relatively much larger than in therian

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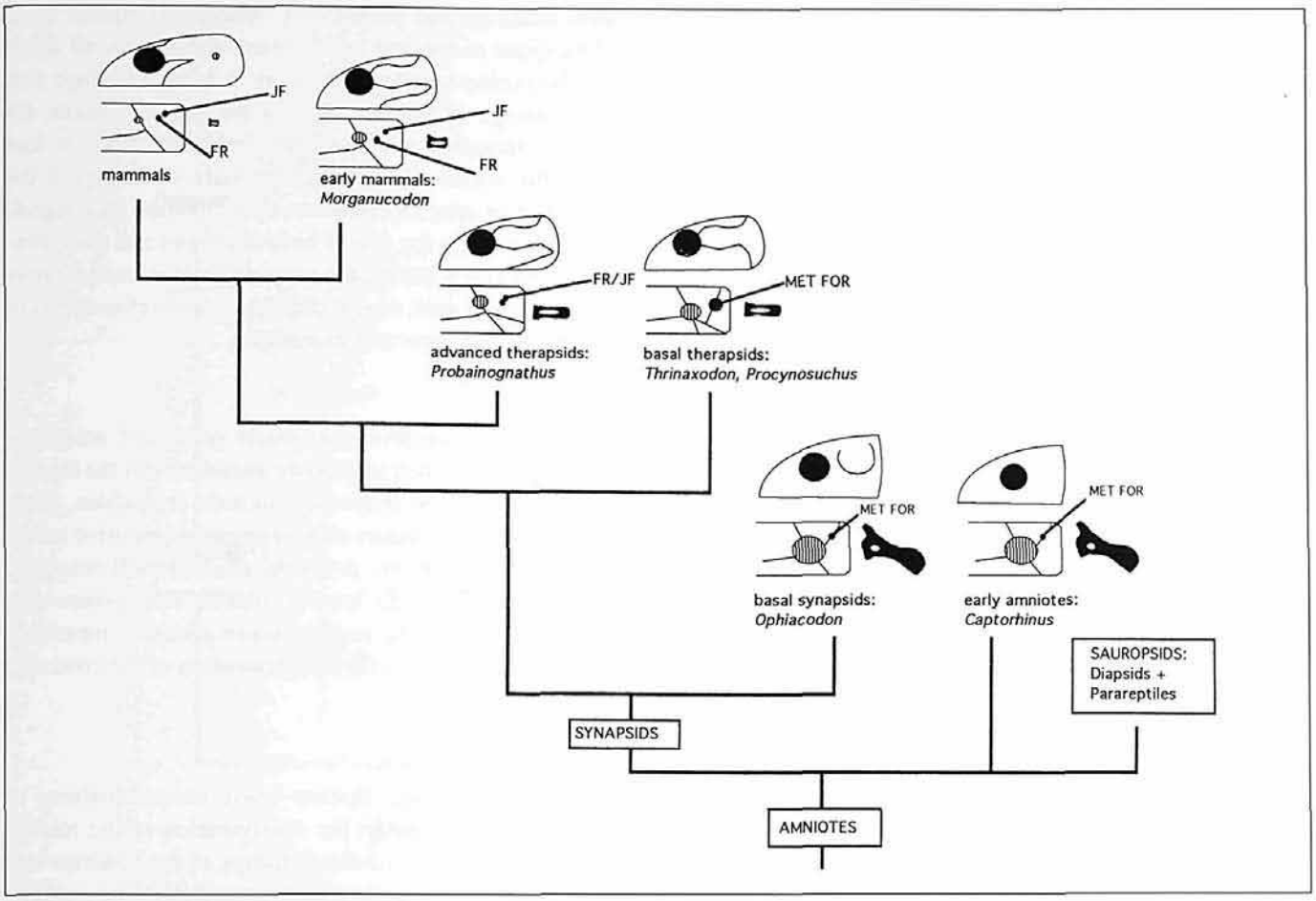


Fig. 3. Cladogram with icons of the synapsid-mammal lineage. Use of the term 'basal' in this figure and elsewhere implies a position at a low branchpoint (node) on the cladogram. The term carries no chronological implications. JF = Jugular foramen; FR = fenestra rotunda.

mammals. The quadrate and articular, which ultimately play the role of incus and malleus, respectively, resemble those of more primitive nonmammalian cynodonts, though the configuration of the other postdentary bones suggest that a tympanic membrane was present. Theoretical studies have analyzed the possibilities for hearing in *Morganucodon;* they have suggested maximum sensitivities to sounds about 6 to 30 kHz, indicating that compared to rodents, it was relatively insensitive to sound [Rosowski, 1992].

Morganucodon is placed by Allin and Hopson [1992] in the same lineage as the somewhat more primitive *Probainognathus.* This animal resembles *Morganucodon* in the configuration of the stapes and postdentary bones, but the round window and jugular foramen are almost confluent, separated only by a small process formed from the opisthotic region of the capsule. *Thrinaxodon* and *Procynosuchus* are successively more plesiomorphic nonmammalian cynodonts (fig. 3) and display the cynodont condition of quadrate and articular; in addition to having the round window and jugular foramen confluent, however, they have successively more primitive states for the fenestra vestibulae. In *Thrinaxodon,* the fenestra vestibuli margin includes the parasphenoid, but the basioccipital is just excluded by the opisthotic [pers. observ., University Museum of Zoology, Cambridge, specimen T 815]. In *Procynosuchus* both basioccipital and the parasphenoid contribute [Allin and Hopson, 1992], making the condition in *Procynosuchus* essentially the primitive one. In these two animals, the postdentary bones remain quite large, and the angular bone forms a reflected lamina as in many nonmammalian synapsids. This structure is thought to be ultimately converted into the mammalian tympanic bone [Allin and Hopson, 1992].

The earliest members of the synapsid lineage, formerly called 'pelycosaurs', include well-known members such as

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Dimetrodon and *Ophiacodon* from the Lower Permian of Texas and New Mexico (fig. 3). Neither of these creatures shows any evidence of specialization of the otic region for reception of air-borne sound. The stapes was a massive strut, whose distal end lodged into a facet in the quadrate [Romer and Price, 1940]. Its precise function is debatable, but it is plausibly interpreted as a bracing bar maintaining the spatial position of the quadrate, which still formed the main jaw articulation. In early tetrapods, it is thought that the flattened stapes, resting against the palatoquadrate, had a similar mechanical function [Carroll, 1980; Smithson, 1982].

The proximal end of the 'pelycosaur' stapes has a large, broad footplate whose rim contacts the otic capsule bones dorsally and the parasphenoid and basioccipital ventrally. The fenestra vestibuli was a large hole whose margins lay on the line of the embryonic metotic fissure, just as in the earliest known tetrapods. A small foramen, assumed to be the metotic foramen, presumably allowed passage of the vagus nerve and jugular vein, but its small size suggests that there was no outlet there for the perilymphatic duct. Internally, the otic capsule was poorly ossified, there being no medial wall separating the otic capsule from the brain cavity and no bony walls defining separate chambers of the inner ear. Also absent were any features suggestive of the presence of a defined middle ear cavity or of a tympanic membrane, there being neither an 'otic notch' nor modifications of the lower jaw bones such as those seen in cynodonts. Essentially, the situation was that of a primitive, early tetrapod.

This primitive configuration is manifested by all of the earliest known amniotes in which the material permits assessment. Early amniotes such as *Protorothyris* [Clark and Carroll, 1973], the earliest known synapsid, *Archaeotyris,* from the Carboniferous of Nova Scotia [Reisz, 1972], the earliest known diapsid, *Petrolacosaurus,* from the Pennsylvanian of Kansas [Reisz, 1977, 1981], and the diapsid *Araeoscelis* from the Lower Permian of Texas [Vaughn, 1955] are all essentially similar in the otic region and can be compared very closely with the well-known early amniote *Captorhinus* from the Lower Permian of Texas, whose otic structure is described in great detail [Heaton, 1979].

In summary, we can trace the development of the characteristically mammalian otic region from its origin among early amniotes and throughout the evolution of the synapsid lineage. Early synapsids maintained an essentially primitive condition of all components, the major changes to the ear region occurring during the Triassic period among cynodonts and early mammals. In cynodonts, changes to the stapes, quadrate and postdentary bones, consequent upon modifications to the jaw mechanism, resulted in all these bones becoming smaller in concert. It is at this stage that the beginnings of features such as the round window are first seen, though conditions then remained more or less stable until, within the earliest mammals, evolution of the round window appears complete, accompanied by a significant diminution in the size of both the stapes and the fenestra vestibuli. It is a reasonable supposition that each of these stages correlates with related changes to and refinements of the inner ear receptors and processors.

The Diapsids

The diapsids comprise two major groups of amniotes: the archosaurs and their immediate ancestors and the lepidosaurs and theirs. The archosaurs include crocodiles, dinosaurs, birds and pterosaurs as well as some primitive members whose phylogenetic positions are less well resolved. The lepidosaurs include lizards, snakes, amphisbaenians, sphenodontids and some poorly known primitive members. I shall start this survey with an examination of the crocodile lineage.

The Crocodile Lineage

Modern crocodilians display many unique features of the otic region, including: the configuration of the middle ear cavity; the anatomical relationships of the quadrate and quadratojugal to the squamosal, braincase and otic capsule; the disposition of the fenestra vestibuli (FV) and pressure relief window (PRW); and the ossifications of the inner ear chambers. The cochlea in crocodiles has a characteristic 'knee-bend' or geniculation, which is reflected in the bony structure of the otic capsule. Crocodiles also have a complex suite of chambers and passages penetrating the braincase, associated with their unique eustachian system. Potentially therefore, the ear region of crocodiles should be as amenable to fossilization as that of mammals, and its evolutionary story similarly interpretable. It has not received nearly as much attention as that of mammals, presumably because people are mammals and not crocodiles, but one or two detailed studies have been made of certain very well preserved taxa in the context of the evolutionary origin of birds [e.g., Walker, 1972, 1990].

Another possible reason for this lack of attention is the controversy over the phylogeny of the crocodile lineage [e.g., Parrish, 1993]. The sequence of acquisition of characters must remain uncertain while the phylogeny remains unresolved, though some general statements can be made. The Crocodylia are a closely related group whose otic structure is more or less homogeneous throughout. Accord-

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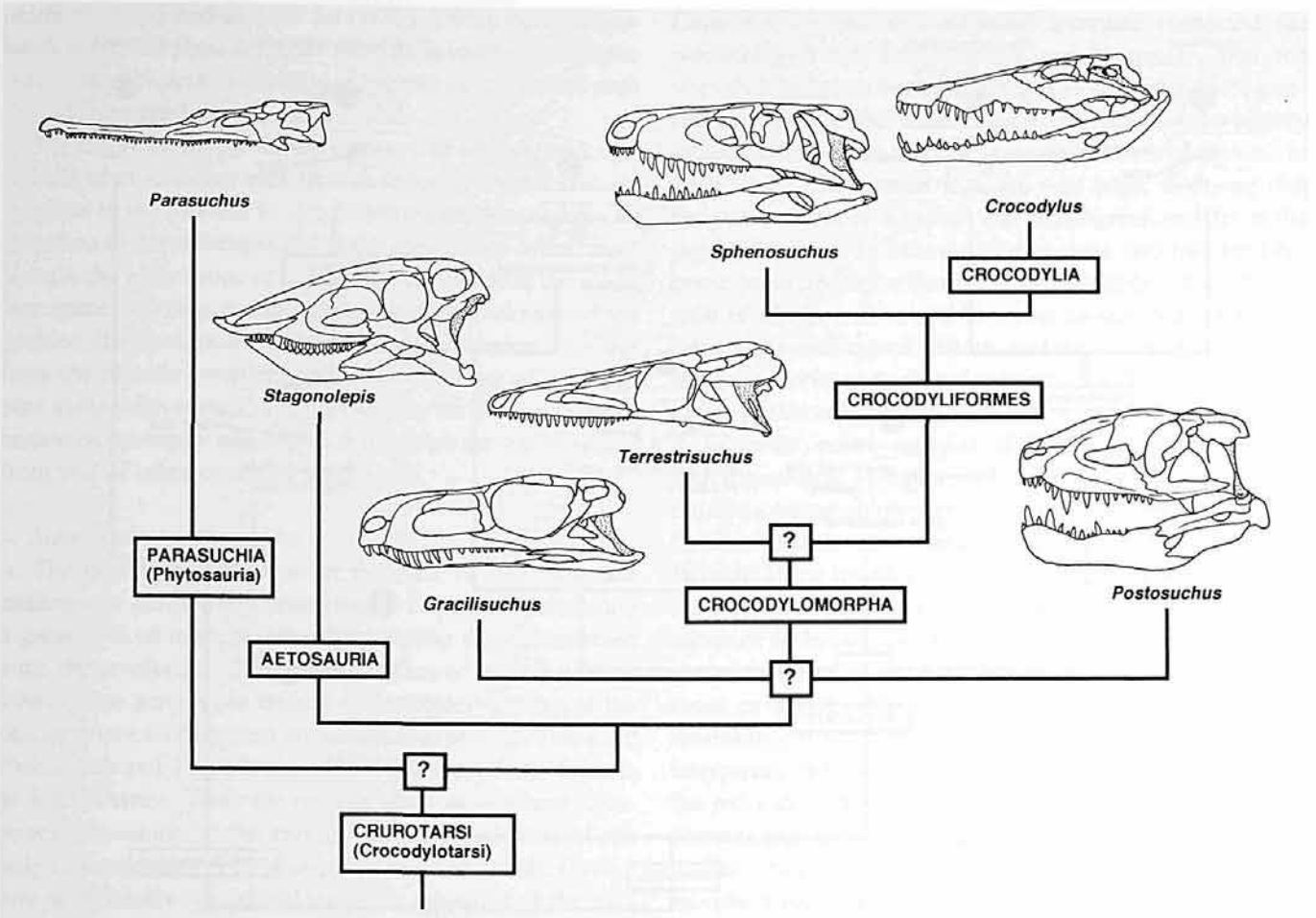


Fig. 4. Cladogram of crododile lineage with skulls illustrated. Quadrates stippled.

ing to some phylogenies (fig. 4), forms such as *Sphenosuchus* [Walker, 1990] lie just outside the Crocodylia among the Crocodyliformes. This genus from the Upper Triassic of South Africa is represented by excellently preserved material which allows details of the otic capsule relationships and inner ear ossifications to be described. For example, most of the complex eustachian system is in place, and the FV and PRW have their characteristic crocodile-type relationships, separated from each other by a narrow curving process of the opisthotic known as the interfenestral bar. As in modern crocodilians, the quadrate forms a complex with the quadratojugal which sutures to the otic capsule and squamosal. This arrangement encloses a narrow V-shaped slot which houses the tympanic membrane. The squamosal provides an overhanging ledge on which the upper 'ear-lid' is supported in modern aquatic forms. However, the elongation of the cochlea and associated geniculation had not yet occurred. It appears that although much of the otic structure associated with the crocodilians had evolved in these Triassic forms, some of the specializations of the inner ear had not.

Primitive members of the crocodile lineage are also known from good material. *Terrestrisuchus*, a late Triassic form from Wales [Crush, 1984], nests within the Crocodylomorpha according to some phylogenies (fig. 4), though others place it with *Sphenosuchus* in a sister clade to Crocodyliformes. Otically, it appears less crocodilian than *Sphenosuchus*. It has a similar arrangement to *Sphenosuchus* of the quadrate, quadratojugal and squamosal, though the V-shaped space they enclose is broader. A ridge running along the dorsal margin of the quadrate is said to have housed the tympanic membrane, though if so, it must have

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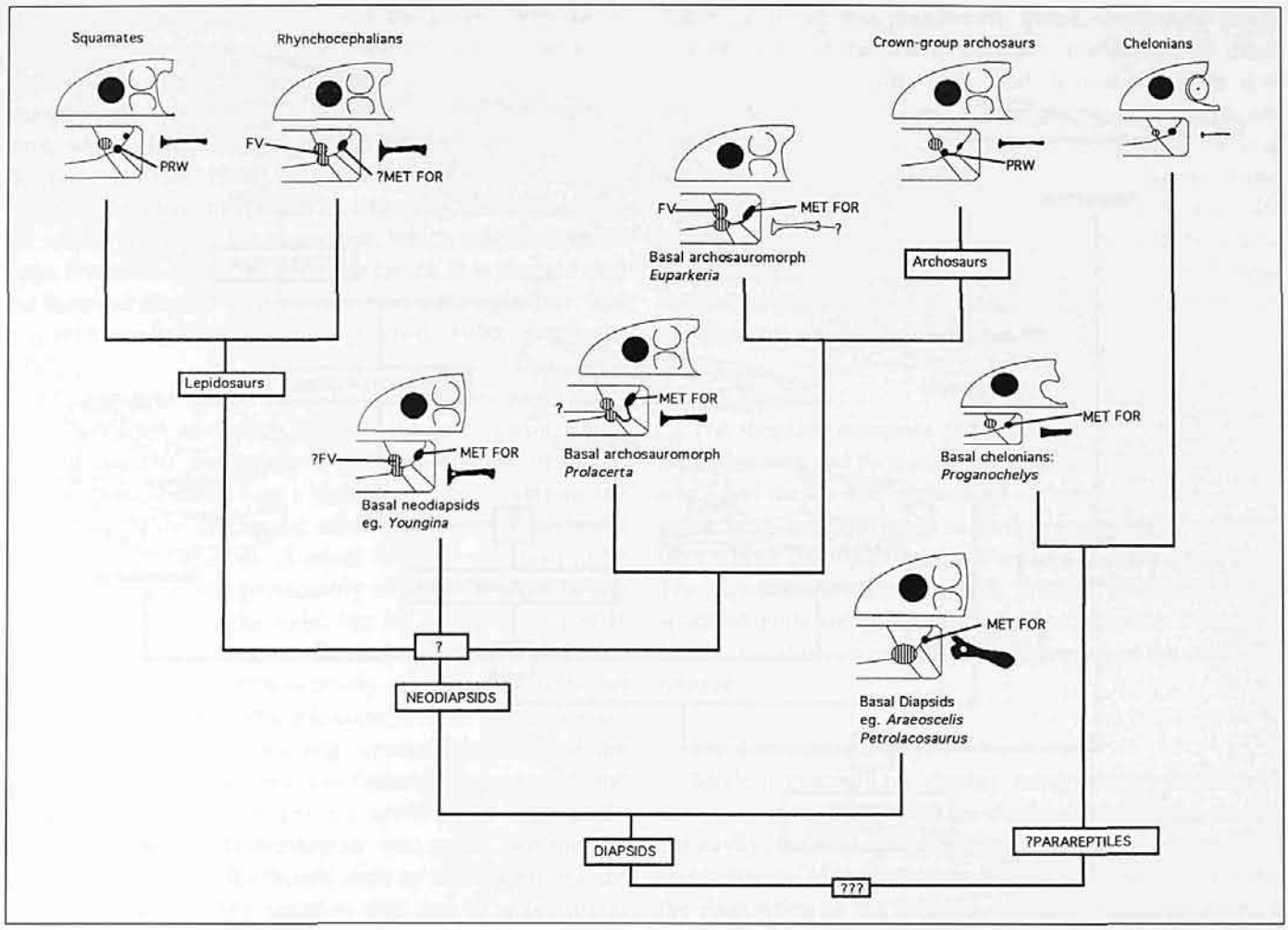


Fig. 5. Cladogram with icons of selected diapsid and parareptile amniotes.

been a rather larger one than in modern forms, and it could not have been protected in the same way by the upper and lower 'ear-lids' which they possess. However, the postcranial skeletons of both *Terrestrisuchus* and *Sphenosuchus* [Walker, 1990] are those of cursorial, terrestrially adapted forms, so that the 'ear-lids' associated with aquatic life may not have been necessary. Nonetheless, it is clear that many of the otic specializations of crocodilians evolved before they became secondarily aquatic. The FV of *Terrestrisuchus* was separated from the PRW ('fenestra pseudorotunda' [Crush, 1984]) by the interfenestral bar, but this foramen was apparently divided in turn by a narrow bar separating what Crush interprets as a foramen perilymphaticum and a perilymphatic duct, the latter confluent medially with the metotic foramen. Much of the internal structure of the otic capsule of *Terrestrisuchus* is not well represented in the fossils, but the quality of that which exists suggests that the original ossification, not subsequent preservation, was poor.

At the base of the crocodile lineage lie many forms whose relationships to each other are uncertain. They include forms such as *Gracilisuchus* and *Postosuchus* [Parrish, 1993] which show otic regions that are successively more generalized and which resemble other early archosaurs more than they do crocodilians (fig. 4). Some phylogenies unite them with phytosaurs and aëtosaurs into a stem-crocodile-lineage called Crurotarsi [Juul, 1994], but none of these taxa shows much evidence of crocodilelike specializations. In many, the otic region is poorly ossified, so that details are not available, but this in itself suggests that crocodilian specializations were absent. In others, for example in the aëtosaur *Longosuchus* [Parrish, 1994], the fe-

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nestra vestibuli and metotic foramen are only barely separated, while the pressure relief window is confluent with the latter. Internal details of the otic capsule or its medial wall are not preserved.

We can trace the evolution of many of the otic specializations of crocodilian ears from a series of Upper Triassic relatives to the modern forms, following the increase in ossification of the otic capsule and the appearance of its medial wall, the elaboration of the bony walls defining the cochlear space and thus the elongation and geniculation of the cochlea, the evolution of a pressure relief window separate from the metotic foramen, and the appearance of the complex eustachian system. In successively earlier forms it becomes increasingly difficult to distinguish the ear structure from that of other primitive archosaurs.

Stem Archosauromorphs

The phylogeny of forms at the base of the archosaur radiation is particularly contentious, but is now attracting a great deal of interest, especially among those concerned with the evolution of braincases [Gower and Sennikov, 1996]. Two genera are critical to the understanding of the otic structure among stem archosauromorph taxa. These are *Prolacerta* and *Euparkeria,* both from the Lower Triassic of South Africa. Their ear regions show us the bony components common to the ears of ancestral members of not only crocodylomorphs but also dinosaurs and birds. *Prolacerta* is generally considered the more primitive of the two, and both are represented by very good material of the braincase and otic region.

In both, the quadrate is shallowly embayed, a condition which has usually been interpreted as indicating the presence of a tympanic membrane. In *Prolacerta* a small pocket on the quadrate's mesial surface may represent the location of a middle ear cavity. The stapes of *Euparkeria* is unknown, but that of *Prolacerta* is slender and has lost its stapedial foramen, as in all later archosaurs where it is known. However, despite the good detail shown by some parts of the otic capsules, there is no medial wall ossified in either genus. Internally, some structures, notably the semicircular canals and anterior ampullary recess, are interpretable where low walls define several pockets and channels, though the interpretation of other structures, such as the facial foramen [Evans, 1986; Walker, 1990] and the lagenar recess, are open to dispute. Significantly, in both genera the rim of the fenestra vestibuli is poorly defined; as preserved, the ventral margin is incomplete. Delicate processes of the opisthotic and proötic probably just contacted each other, as suggested by broken remnants. An unossified gap beneath corresponds to the region where in early amniotes, such as *Captorhinus,* the large stapedial footplate contacted the parasphenoid and basioccipital, and it appears that the stapedial footplate became reduced in diameter among archosaurs before the fenestra vestibuli became the tightly defined foramen in the otic capsule of later amniotes. In both forms, the metotic foramen was large, implying that the pressure relief window was undifferentiated from the jugular foramen. In later archosaurs these two foramina become separated by a flange of the opisthotic. The otic region of *Prolacerta* resembles most closely that of *Sphenodon* in its ossification pattern, and the size and disposition of the metotic and other foramina. The potential significance of this will be discussed below.

In recent years, material of *Prolacerta* [Evans, 1986] and *Euparkeria* [Welman, 1995; D.J. Gower, pers. commun.] is being studied anew. Existing material, prepared further, reveals new details [*Prolacerta,* pers. observ., Bernard Price Institute, specimen 2675]. Reinterpretations are bound to result, especially in the light of revised assessments of archosauromorph phylogeny. Welman [1995] suggested that *Euparkeria* has a braincase and otic structure far more primitive than that of dinosaurs but nevertheless resembling that of birds closely in several characters. His interpretation is at odds with those that place birds among the more derived dinosaurs [e.g., Chiappe, 1995] and will almost certainly be challenged in the future.

The rhynchosaurs are a group of basal archosauromorphs which were once united with the sphenodontians in the Rhynchocephalia [Romer, 1966] on the basis of dental characters. However, this interpretation is no longer supported, and sphenodontians (rhynchocephalians) are now considered to be early lepidosauromorphs. Most rhynchosaurs were moderately sized vegetarians with highly specialized dentition and no obvious otic capsule specializations, no tympanum and no middle ear cavity. However, *Mesosuchus,* from the Lower Triassic of South Africa, considered to be the most primitive member of the group [Dilkes, 1995], has not been described in detail. Its otic structure is extremely well preserved [pers. observ., South African Museum, specimen 6536] and could form a test for the ancestral otic conditions of archosauromorphs inferred from *Prolacerta* and *Euparkeria.*

Dinosaurs and Birds

The otic region has been described in a number of dinosaurs, though not in the context of ear evolution. This could form an illuminating study, especially, for example, among hadrosaurs, whose peculiar cranial appendages are thought to have been employed in sound production. The implication is that they had acute hearing, capable of discriminat-

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ing intraspecific sounds. It is known that many dinosaurs had such features of the inner ear as a separate PRW, and a delicate stapes. However, it is also known that the earliest dinosaurs had ear regions that were essentially like those of stem archosauromorphs [e.g., *Lesothosaurus,* Sereno, 1991] without such features as a separate PRW, which makes one question their hearing capabilities.

Archaeopteryx, considered to be the earliest known bird, has a relatively poorly known otic region, though Walker [1985] has described birdlike features. The PRW shared a common opening with nerve IX according to his interpretation. Welman [1995] does not consider that *Archaeopteryx* shows any special similarity to dinosaurs in the otic region; rather, it is more primitive than the theropods to which it is supposed to be most closely related. The interrelationships of birds and dinosaurs are currently the subject of intense debate as a result of a flurry of new finds [Chiappe, 1995; Hou et al., 1996], so that the early evolution of bird ears and hearing is a study for the future. *Archaeopteryx* also shows birdlike tympanic pneumatization which is different from that in crocodiles and so is probably a convergent feature [Witmer, 1990]. Similarities that Walker [1990] noted between *Sphenosuchus* and birds, and which also exist in their soft tissue structures, may be general archosaur synapomorphies rather than bird-crocodile ones [Carr, 1992], but this whole area is in need of restudy and no more detailed analysis will be attempted here.

Lepidosauromorphs

Among modern lepidosaurs, only lizards exhibit specializations associated with a tympanic ear. It is 'the lizard ear' that is often taken as a model for that of 'reptiles' [e.g., Romer, 1962] as a whole, though their inner ear anatomy, in features such as the basilar papilla hair-cell orientation and the structure of the tectorial membrane, show a bewildering amount of familial diversity [Wever, 1978; Miller, 1992]. It is not obvious what this diversity signifies, though it has been suggested that it may indicate a lack of strong selection pressure on a rather 'unimportant' system, or it may be that lizard families developed specialized receptor structures separately, starting from a similar plesiomorphic baseline [Köppl and Manley, 1992; Miller, 1992].

The fossil record is as yet little help in determining the evolutionary history of lizard ears. The earliest members of modern groups are known from the Middle and Upper Jurassic but most are fragmentary and few include suitable braincase material [Evans, 1993]. Potentially however the evolution of the ears among modern families is traceable.

The most interesting modern lepidosaur from a phylogenetic and evolutionary point of view is the rhynchocephalian *Sphenodon.* It seems to show the plesiomorphic state of many otic features in its skeletal and soft tissue anatomy: for example, in its lack of a separate pressure relief window, lack of ossification of the medial wall of the otic capsule, and barely completed ventral margin of the fenestra vestibuli. Most notably, it lacks a tympanic membrane and middle ear cavity. Some have considered this a primitive absence, though Wever [1978] regarded it as a secondary loss, based on the fossil record of rhynchocephalians. Recently described fossil rhynchocephalian material is relevant here.

Rhynchocephalians are the most basal group of the lepidosaur crown clade, known from an abundant fossil record of Triassic forms. *Clevosaurus* is a derived rhynchocephalian from the Upper Triassic. Although Wu [1994] described a separate PRW in the braincases of *C. wangi* from China, S.E. Evans [pers. commun.] disagrees and interprets it as having an undivided metotic fissure. However, it does have a tympanic quadrate conch very like that of squamates. Wu proposed the configuration of the middle ear as a synapomorphy uniting squamates and rhynchocephalians, supporting Wever's opinion that the middle ear of *Sphenodon* is 'degenerate'. The braincase of the most primitive rhynchocephalian, *Gephyrosaurus,* is currently under study [S.E. Evans, pers. commun.]. If *Gephyrosaurus* has a separate PRW, we may be better justified in regarding *Sphenodon* as secondarily tympanum-less. Meanwhile, because of its arguably plesiomorphic inner ear anatomy and, additionally, in its similarity in bony structure to stem archosauromorphs, it may still provide a valid model for the otic condition and auditory capabilities of basal neodiapsids (lepidosauromorphs plus archosauromorphs).

In lepidosaurs, a tympanic ear is always found in association with a distinct pressure relief window, though both may be secondarily lost as apparently in chamaeleons [Wever, 1978]. The presence of a PRW may plausibly be suggested as indicative of the presence of a tympanic ear in squamates. Significantly, the formation of the PRW from part of the metotic fissure is different in detail in lepidosaurs from that in archosaurs, and it bears a different relationship to the opisthotic, exoccipital, and basioccipital bones. This region of the braincase in diapsids is being studied afresh [S.E. Evans, D. Gower, pers. commun., work in progress], but already a case can be made for independent PRW evolution. This suggests that the fully formed tympanic ear may also have arisen separately in these two diapsid groups. Studies of the fine detail and embryology of the soft tissue structures in each and the differences that they may show could test this hypothesis, but it may also be tested by examination of the most primitive neodiapsid forms.

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Youngina capensis

This form, from the Upper Permian of South Africa, is critical to the phylogeny of early neodiapsids. Some authors have placed it as a stem lepidosauromorph [Evans, 1988; Rieppel, 1994], while others place it prior to the split between the lepidosauromorphs and archosauromorphs [Laurin, 1991; Juul, 1994]. From the point of view of otic evolution, its importance lies in showing no evidence of a quadrate emargination for a tympanum or middle ear cavity [Gow, 1975], and though the stapes is relatively slender, the footplate is relatively large, and as currently understood [Gow, 1975; Evans, 1987], the margin of the fenestra vestibuli includes the parasphenoid. Further study may change this impression, but it is nonetheless apparently primitive in most describable otic characters and shows poorly defined internal ossifications [Evans, 1987]. If *Y. capensis* is either a stem lepidosauromorph or a stem archosauromorph, its condition would strongly support independent evolution of a tympanic ear in those two groups. If it is primitive to both, it may give a clue to the ancestral conditions of each, though only further information from somewhat more derived neodiapsids could clearly resolve the issue.

Taking diapsids as a whole, I suggest that the role of the stapes, as in synapsids, may initially have been as a structural skull member. Release of the stapes from this role may have been facilitated by the development of the characteristic neodiapsid elongate paroccipital process that fastened securely to the squamosal at the posterodorsal corner of the skull. In synapsids, the occipital region of the skull became consolidated in a comparable but different way.

Turtles

As with mammals and crocodiles, so turtles (chelonians) also have highly ossified otic regions and a multitude of unique otic characters that are potentially fossilizable. These include the cone-shaped excavation of the quadrate for the tympanic membrane and middle ear cavity, and the ossifications surrounding their apomorphic reentrant fluid pressure relief system. The story of turtle ear evolution should be relatively easy to trace, at least from the earliest known turtle, *Proganochelys* from the Upper Triassic of Germany, through to modern forms. *Proganochelys* shows little in the way of development of any of the turtle ear specializations, but instead resembles that of a basal neodiapsid in many respects [Gaffney, 1990]. It has a shallowly embayed quadrate, a relatively stout, imperforate stapes, with a fenestra vestibuli that is larger than that of modern forms but nevertheless shows a derived condition in being positioned entirely within the otic capsule. The problem attaching to turtle ear evolution lies in the debate over chelonian relationships to other amniotes.

Although almost universally agreed to be more closely related to diapsids than to synapsids [Gauthier et al., 1988], debate about the nature of that relationship still rages. Recently, following studies of pareiasaurs and their relatives the procolophonids, there has emerged a fairly widely held view that turtles, along with these fossil forms, are part of a parareptile clade whose base is formed by a group of small creatures called millerettids from the Upper Permian of South Africa [Gow, 1972]. Millerettids show no otic embayment of the quadrate; the stapes is fan-shaped, resembling those of early tetrapods; and the otic capsule shows a plesiomorphic structure. While pareiasaurs show little in the way of quadrate embayments, in procolophonids this feature is often exaggerated. What this says for early turtle otic conditions depends on which of two competing parareptile phylogenies is accepted [Lee, 1995; Laurin and Reisz, 1995]. On the other hand, an even more recent analysis of turtle relationships places them fully within the diapsids [Rieppel and de Braga, 1996], a position that is at least congruent with the bony ear structure of *Proganochelys*. Soft tissue anatomy strongly suggests that turtle ear features originated independently among amniotes [Wever, 1978]. Thus, chelonians may represent a further iteration of the development of a tympanic ear 'from scratch', complete with a unique system of pressure relief, or they may have come equipped with a basic diapsid set of features inside and out. *Sphenodon* may provide a good model for the ear of *Proganochelys* too, but if so, the possibility remains of independent elaboration in pleurodiran and cryptodiran forms.

Amphibians

Arguments for the relationships of modern amphibian groups to each other and to potential fossil relatives are among the most contentious in all tetrapod phylogeny. Very briefly, many people accept lissamphibians (a clade formed by anurans, urodeles and caecilians) as a monophyletic group derived from the fossil temnospondyls [Milner, 1988; Trueb and Cloutier, 1991]. Temnospondyls arose in the late Mississippian and have been interpreted as the earliest group to evolve a tympanic ear [Clack, 1992; Milner and Sequeira, 1994]. This ear structure would thus have been inherited by frogs and lost in urodeles and caecilians. However, controversy surrounds every step of this argument. Some later temnospondyls that were formerly almost universally considered to have had tympanic ears have been

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reinterpreted as possessing, instead, a functional spiracle, at least in the juveniles [Warren and Schroeder, 1995]. The interpretation of other temnospondyls therefore falls into doubt. Caecilians have Lower Jurassic representatives (with limbs) in the fossil record [Jenkins and Walsh, 1993] whose skull structure suggests that they are unlikely to have evolved from any animal with an 'open' structure like that of frogs or salamanders [Carroll and Jenkins, 1996]. Yet other phylogenies have most recently placed modern amphibians in a clade with microsaurs and other Palaeozoic 'lepospondyl' groups, all of which lack otic specializations, while temnospondyls represent an independent basal radiation of tetrapods [Laurin and Reisz, 1997]. Temnospondyls and frogs would thereby represent two separate iterations of the tympanic ear instead of one, the earliest known creature with a froglike tympanic ear then being *Triadobatrachus* from the Lower Triassic of Madagascar [Rage and Roček, 1989]. Until these phylogenies are resolved, the history of frogs' ears must remain a mystery.

Conclusions

This overview of tetrapod otic construction and evolution as seen in the fossil record adds depth to the statement that tympanic ears arose several times independently. It is possible to document the evolution of a number of bony otic characters associated with such a form of ear and several inferences such as those below may be testable in the future.

1. In early tetrapods, there appear to have been no otic specializations. The stapes was a mechanical component of the skull, and the internal bony structure of the otic capsule suggests poorly isolated otic components. It cannot be assumed, given the difference between modern amphibians and amniotes, that the earliest tetrapods had a perilymphatic duct, and given the absence of an auditory papilla or even a separate lagenar pouch in the lungfishes [Fritzsch, 1992], their closest relatives [Cloutier and Ahlberg, 1996], it must be seriously questioned that they had any specialized auditory inner ear receptor. The capability for reception and processing of airborne sound in these predominantly aquatic animals [Clack and Coates, 1995] seems even less likely.

2. The fenestra vestibuli in adult early tetrapods lay between the otic capsule and the occipital arch components of the skull, with a contribution from the parabasisphenoid in all except *Acanthostega*. This large hole becomes both smaller and confined to the otic capsule, independently, in each of several amniote lineages: synapsids/mammals; diapsids, with possibly separate iterations in lepidosauromorphs and archosauromorphs; and (probably) parareptiles/turtles. Whether or not lissamphibians represent a monophyletic group descended from temnospondyls, their FV represents a further instance of this process.

3. Increasing differentiation of the internal ossifications of the otic capsule occurs separately in mammals, lizards, crocodiles and turtles, while the sequence in dinosaurs and birds remains to be studied. It is suggested here that this phenomenon correlates with increasing hearing acuity and the need to isolate the sound reception mechanism from other parts of the body.

4. A distinct pressure relief window for inner ear fluid vibration, or the alternative but equally specialized mechanism found in turtles, may correlate closely with the development of a true tympanic ear. A PRW formed from a remnant of the metotic foramen, or on the line of the embryonic metotic fissure, can be tracked in the fossil record and can be shown to arise independently in mammals, crocodiles, dinosaurs, and lizards. The evolution of a PRW in birds may prove to be yet another separate iteration.

5. On the basis of its bony anatomy and its apparently unspecialized inner ear, *Sphenodon* may make a good model for the auditory capability of many early amniotes, including early neodiapsids, whether or not any Triassic sphenodontids had evolved a fully tympanic ear.

6. The evolution of the FV, PRW, and inner ear ossifications may correlate with stages in the independent evolution of several soft tissue characters of the ear in amniote groups. Plotted onto a cladogram, it may be possible to infer what occurred when and in which groups. It may also be possible to predict differences in soft tissue characters that may be found among the amniote groups in addition to those many which have already been noted [Carr, 1992; Manley and Gleich, 1992]. For example, the tympanic membranes of lizards and mammals form differently embryologically [Presley, 1984]. Crocodile and bird tympana could be checked to test their possible separate evolution from each other or from those of lizards.

7. The fossil record can give information about the timing as well as the sequence of otic character evolution. In all the amniote groups dealt with here, there is a strong signal that the beginnings of otic specializations in each start in the Triassic period (fig. 2). This may represent collector bias or may be a real phenomenon. If real, there may be a number of explanations that can be tested by further investigation. One potentially testable hypothesis is that this timing coincides with the evolution of buzzing or stridulating insects; it is known that the earliest dipterans are found in the Upper Triassic [Fraser et al., 1996]. It is important to

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remember that mammals and birds are the only amniote groups in which there is widespread use of sound for intraspecific communication. I tentatively suggest that the earliest amniotes had auditory capabilities similar to those of *Sphenodon* and that the capacity for hearing (moderately) high frequency airborne sound first arose in amniotes among small insectivores of different lineages. Only later did this facilitate the use of sound for communication. This can be tested by examination of the fossil record of insect groups.

8. Only fossils, in combination with a well supported phylogeny, can document the history of the evolution of the ear. This in turn may suggest new questions to be asked of the fossils themselves and of extant animals, to the enrichment of our understanding.

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